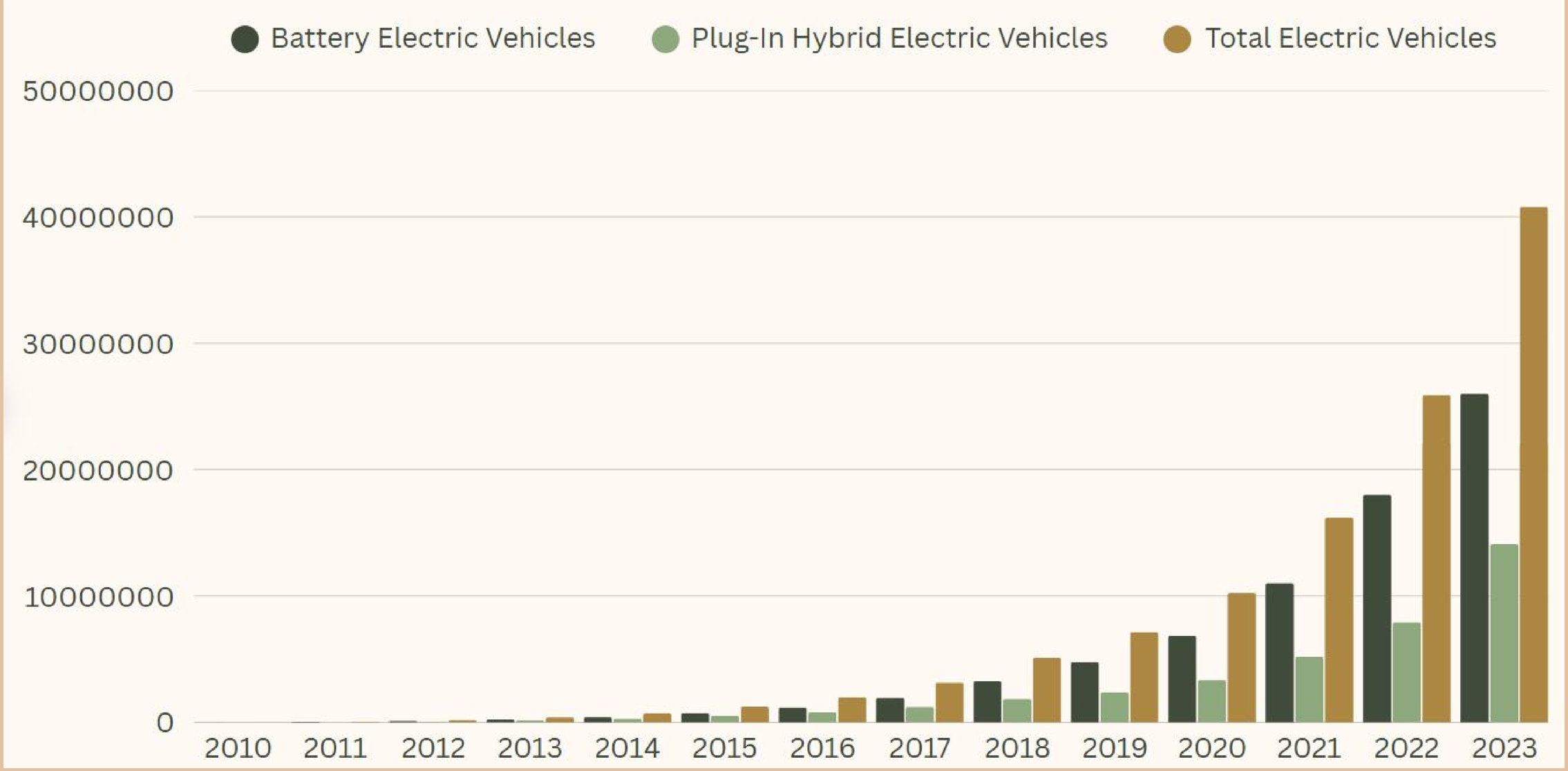
40 million EVs

$500.48 billion valuation

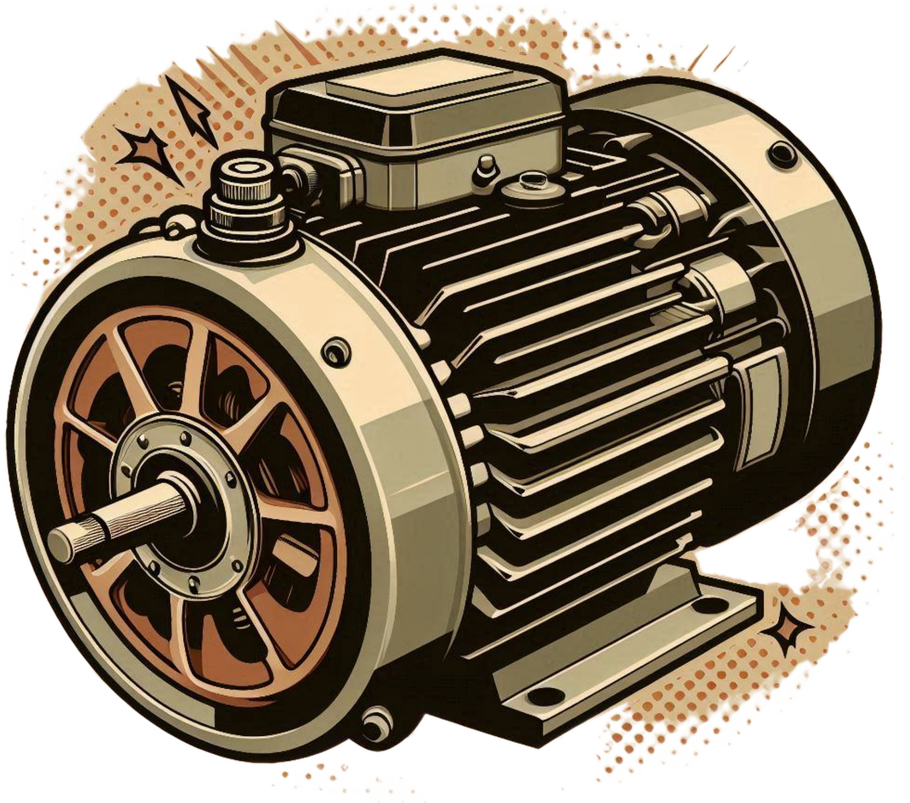
Predicted to reach **$1.5 trillion** in 2030

Source: Statista (2024)





Source: Statista (2024)



At the heart of this growth?

Asynchronous motors

robust, cost-effective, and reliable

The team and my role

- **Role:** UX Lead
- **Timeline:** Aug 2022 - Feb 2024
- **Team members (21):** 1 Director, 2 Sr. Dev Managers, 6 Sr. Devs, 3 Devs, 3 QE, 1 Sr. UX Manager, 2 UX Researchers, 1 UX Lead
- **Constraints:** C++ and Java based infra, legacy code-base, limited GUI components, legacy design system



Challenge – Tuning the Heart of EV

- Asynchronous motors are low maintenance and durable.

- But accurately estimating motor parameters, like resistance, inductance, and inertia, is critical for:
  - Optimizing motor control
  - Maximizing efficiency
  - Reducing development time

- Manual methods and trial-and-error cost time and money.



Solution

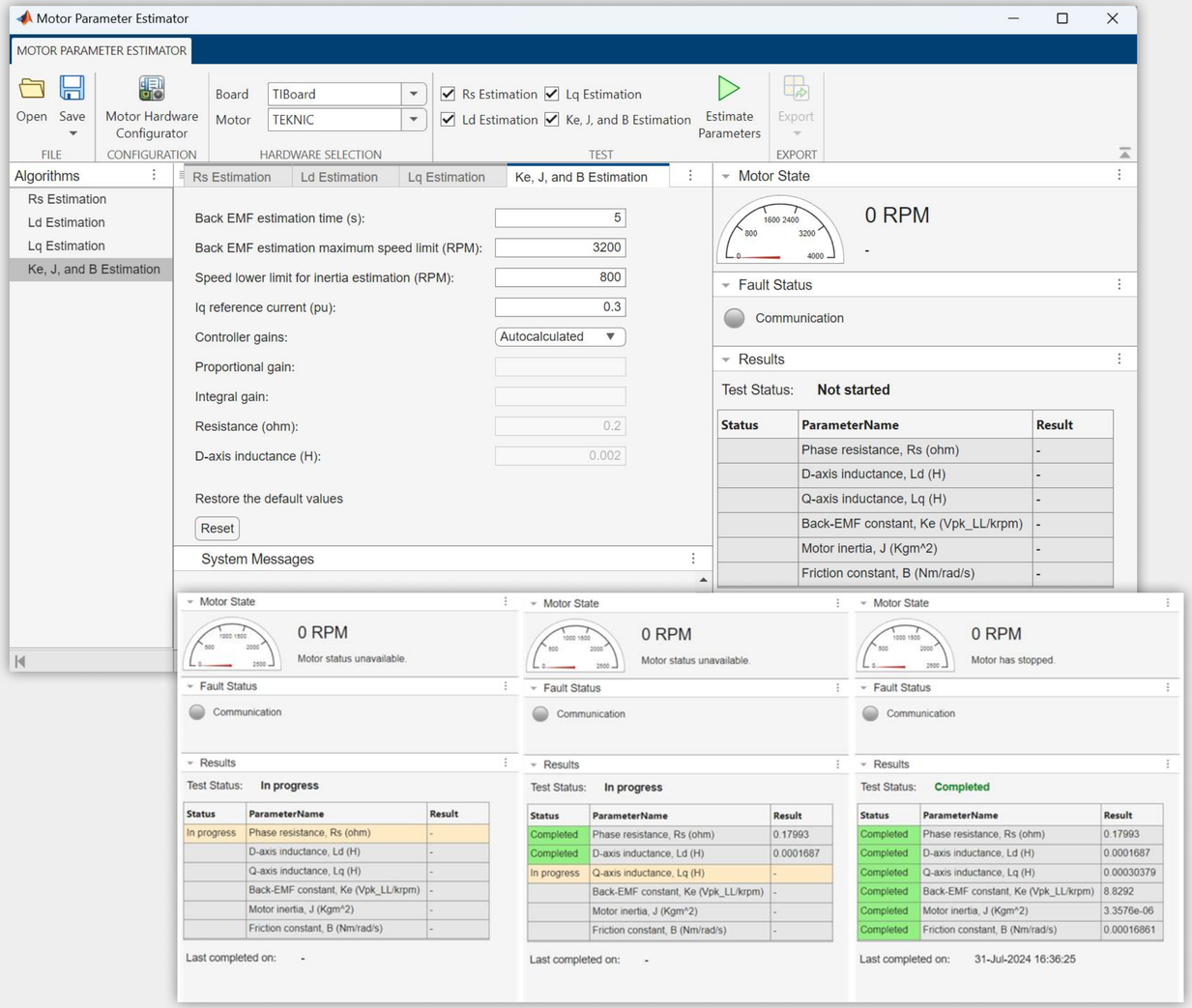
- Motor Parameter Estimator App
  Automate. Simulate. Accelerate.

- Makes motor modeling fast, accurate, and repeatable.

- Features:
  - Estimate parameters

  - Seamless integration with Simulink and Motor Control workflows

  - Enables hardware-in-the-loop (HIL) testing and early-stage simulation



Empowering EV Innovation Across the Globe

- Trusted by global leaders and emerging EV startups alike
- Supports localization, scalability, and compliance with global standards
- The Motor Parameter Estimator App helps Electric OEMs:
  - Slash development cycles
  - Improve energy efficiency
  - Speed up homologation and testing









Gulfstream





Impact Story

Problem Statement

- **I am** an aerospace leader with over 150,000 employees and $50B+ in annual revenue
- **Trying to** introduce electrification in my products
- **So I can** reduce emissions, improve efficiency, and lead the transition to cleaner aerospace technologies
- **But it's difficult to** precisely estimate PMSM parameters which are essential for optimal motor control
- **Which makes me feel** that despite our engineering capabilities, a core technical bottleneck is limiting our ability to scale electrification reliably and efficiently across our product portfolio

Primary OKR

- Electrification
- Sustainable tech

Success Metrics

- Estimation Accuracy
- System Efficiency Gain
- Reduced Development Time
- Flight Test Reliability
- Enhanced responsiveness

Business Impact

- Reduced prototyping and calibration costs
- Improved Energy Utilization
- Sustainability Leadership
- Faster Time-to-Market
- Risk Mitigation

Customer Impact

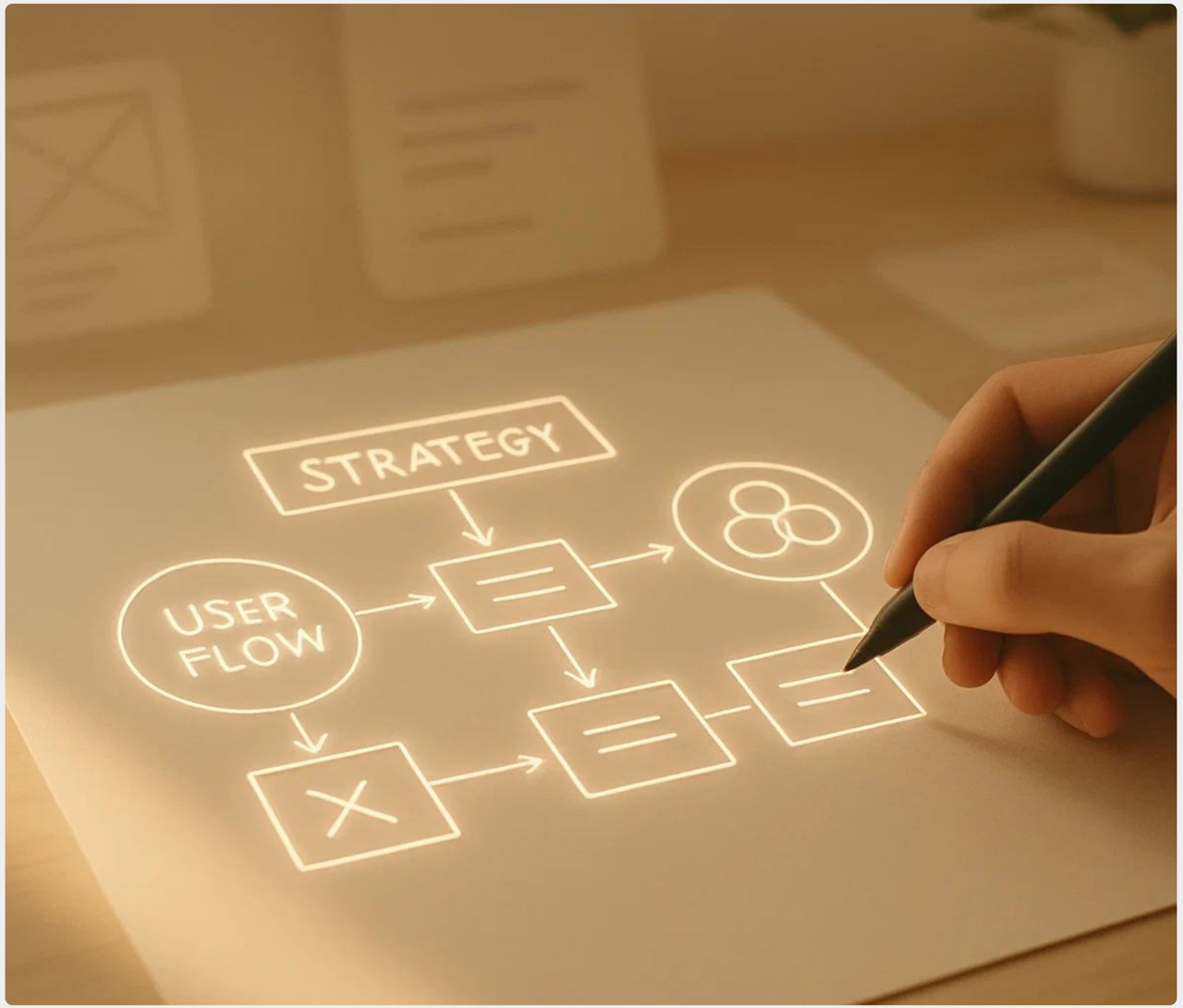
- Increased Aircraft Efficiency
- Fewer maintenance
- Lower Noise Footprint
- Eco-Conscious Branding
- Better control

Design Strategy

Engineering Meets Empathy

Bridging product vision with system-level thinking

- Uncovered friction points using human-centered design
- Shaped features and flows with design thinking
- Aligned UX, tech, and business via systems thinking
- Built clarity with journey maps, personas, and prototypes
- Balanced desirability, feasibility, and viability



The Approach

Discovery

- Customer request patterns & stakeholder interviews
- Market analysis
- Recruitment & primary user interviews
- Thematic analysis → insights & requirement spec

Deliverables

Research summary + prioritized requirements

Design

- Design workshops: prioritization, Information Architecture, rapid prototypes
- Hi-fidelity & coded prototypes; internal design reviews
- Design handover & dev support

Deliverables

Solution concepts + information architecture + prototypes

Reviews

- Research reviews
- Design reviews
- Hardware design reviews (HDR)

Deliverables

Usability report + acceptance checklist



Phase 1

Knowing the User

User roles and goals

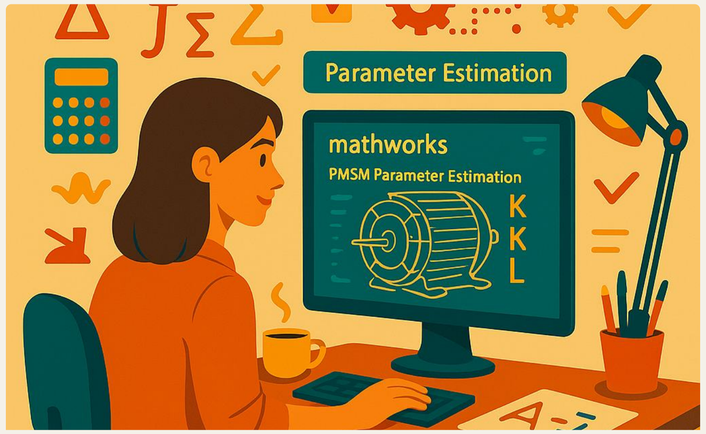
Who are the primary users?



Priority: High

Role: Software Design Engineer

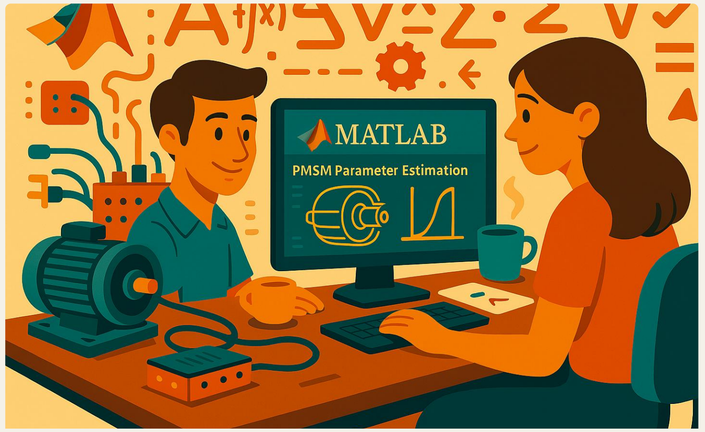
Goal: Create software to control electric two wheeler



Priority: High

Role: Internal developer

Goal: Trying to create tutorials for PMSM estimation



Priority: Medium

Role: Student

Goal: Find PMSM parameter using a hardware

User pains

Pain points from user research

- Various **workflows** of the PMSM estimation process are **scattered**, which makes it difficult for a new user to navigate the workflow
- No control over individual test runs
- In case of PMSM test failure the user has to run the whole test again
- The user does not have access to source code, so it is impossible for the user to understand the estimation process
- Customize shipping examples for various hardware
- ..... few more (not relevant to this app though they affected other app workflows in an integrated system)

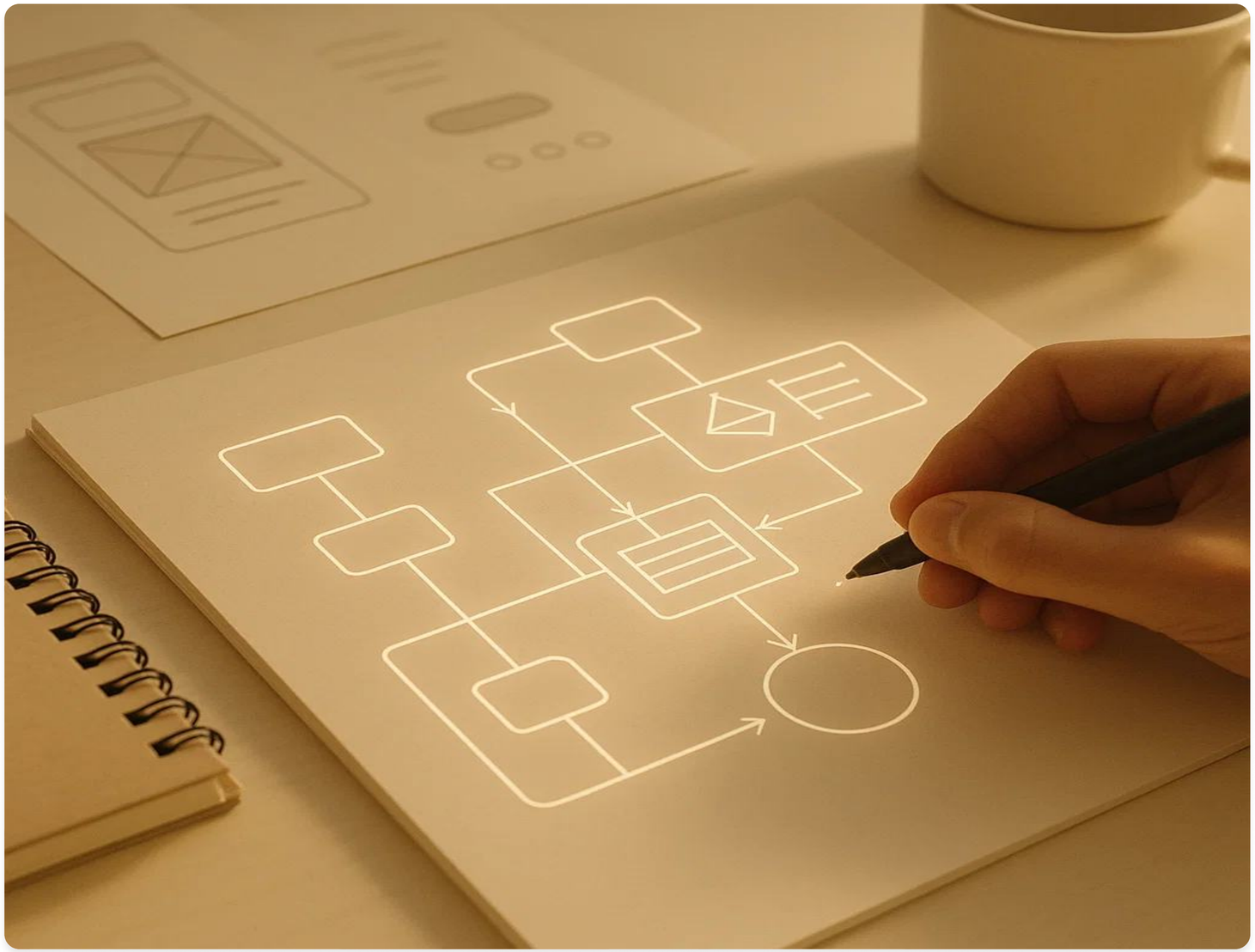


Requirements

Solution towards

- **Intuitive GUI** to simplify PMSM parameter estimation and reduce scripting overhead
- **Flexible test configuration**, including full or selective test runs with status indicators
- **Safety-first design**, including emergency stop and feature-lock during test execution
- **Session management** with options to save progress and export results
- **Real-time feedback**, including motor running status and completion warnings





Phase 2

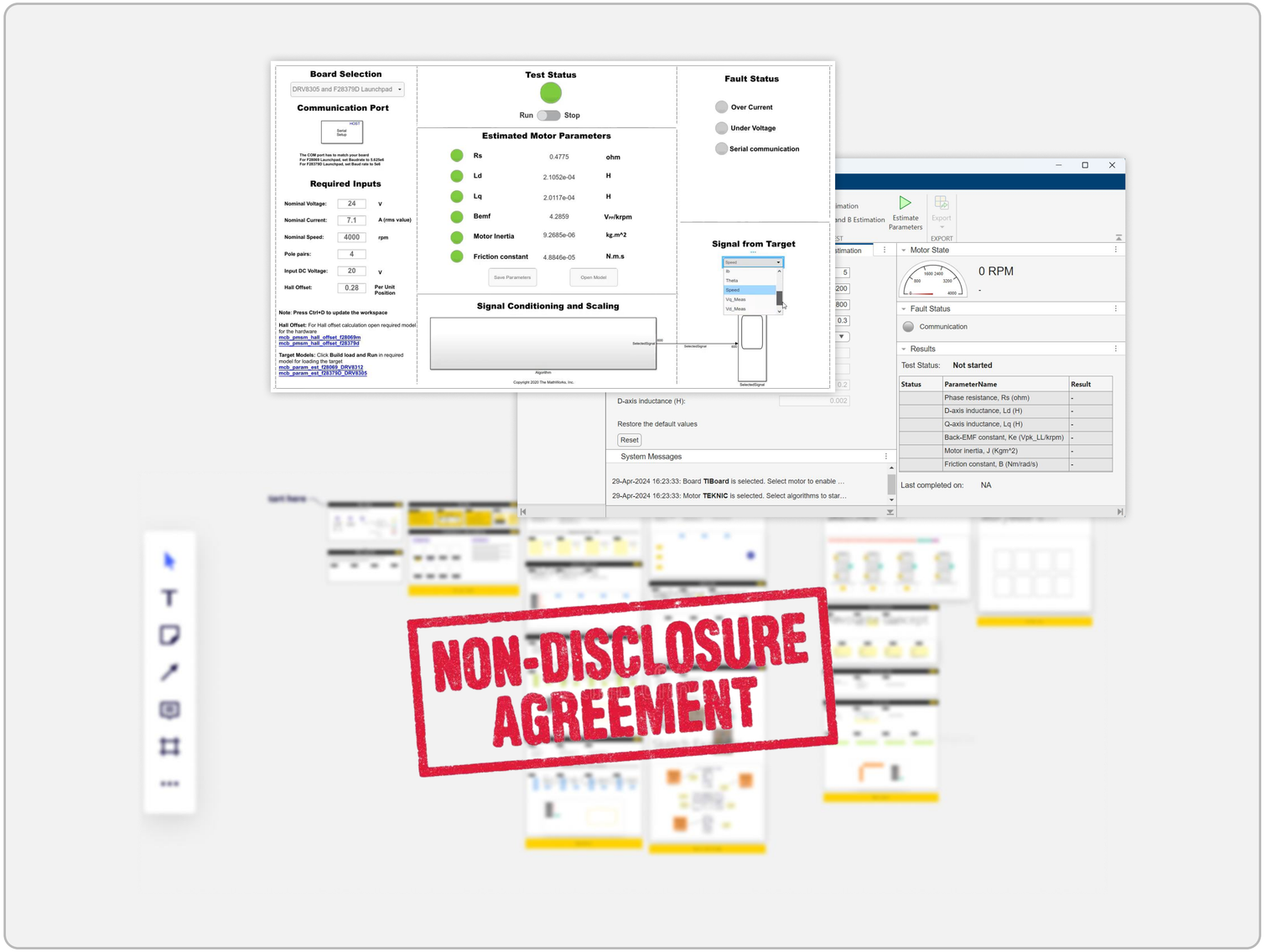
Design

Design Workshop

Direction setting brainstorming
Simulink/Toolstrip App/API

App design workshop
Toolstrip app

- Two sessions, 3hrs each
- 20+ participants
- 1 facilitator, 1 notetaker
- 100+ ideas
- 3 final concepts

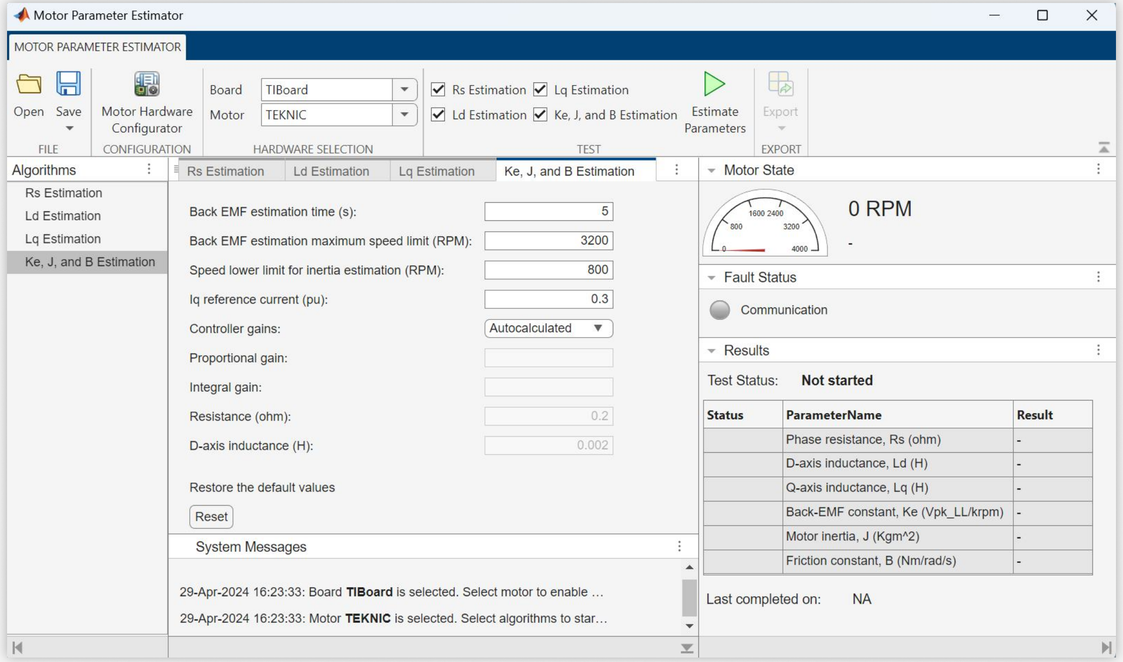


Design Output

Motor parameter estimator application

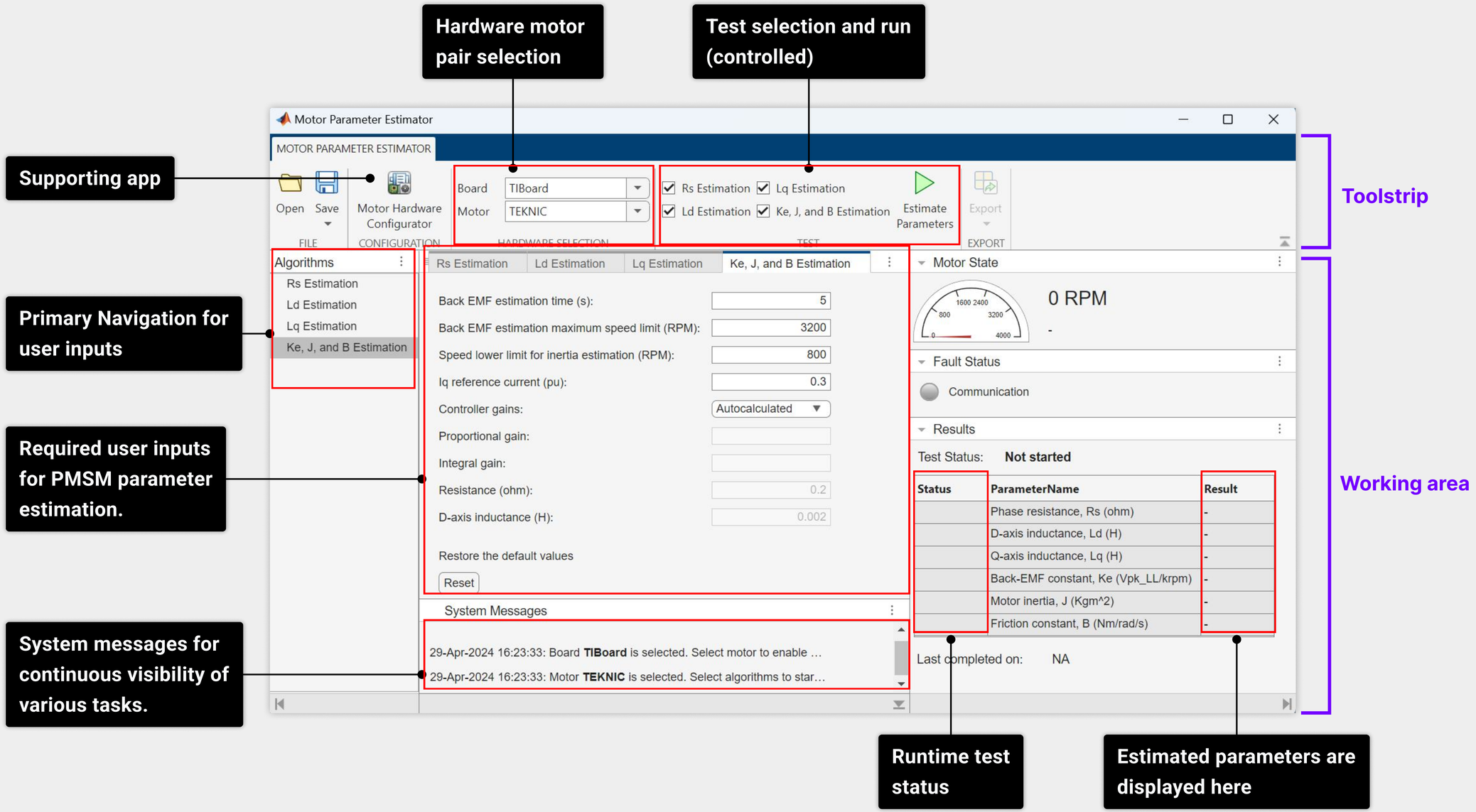


Before - **Simulink model**



After - **Web Application**

Design

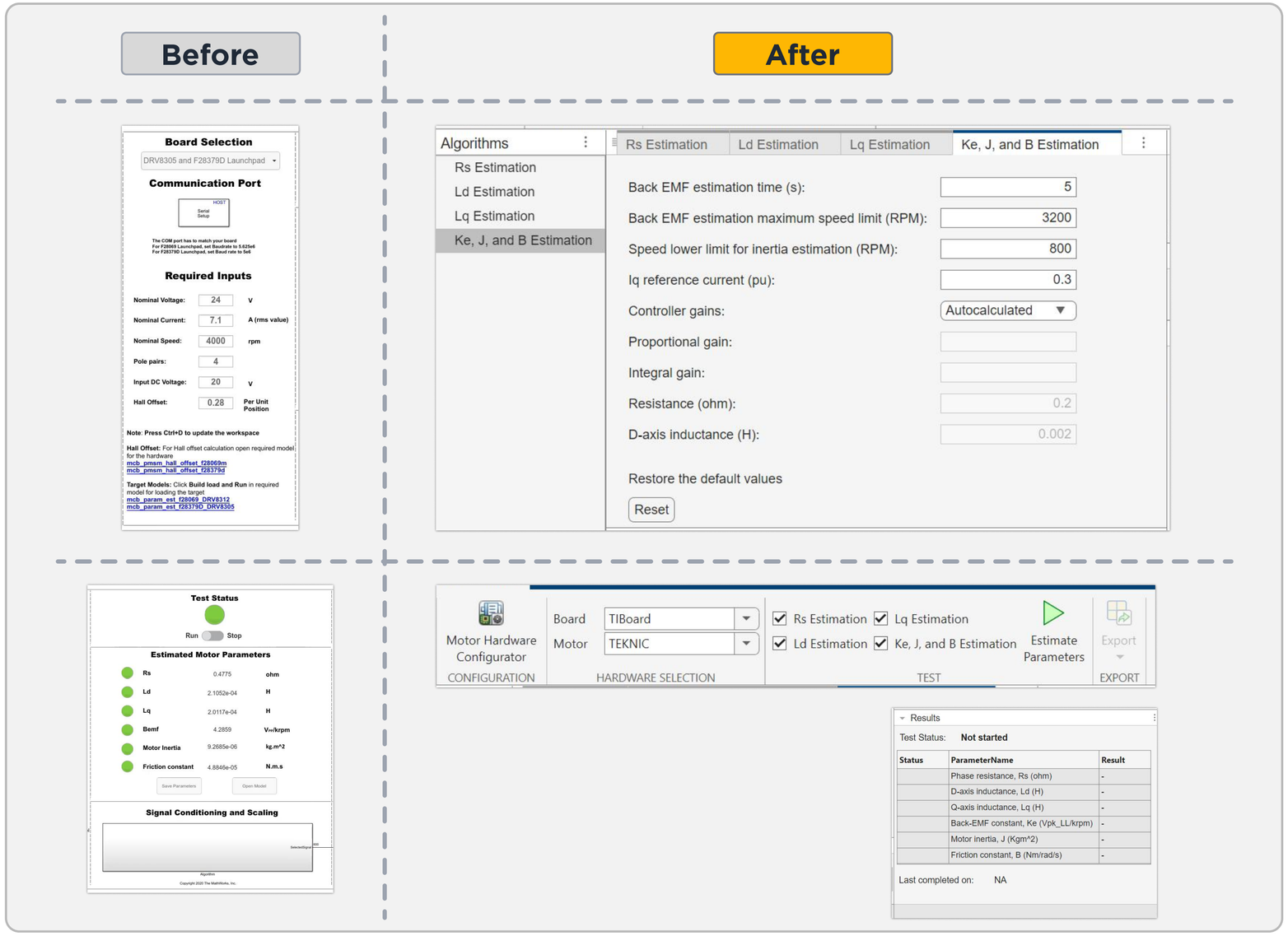

Hardware motor pair selection
Test selection and run (controlled)
Supporting app
Primary Navigation for user inputs
Required user inputs for PMSM parameter estimation.
System messages for continuous visibility of various tasks.
Toolstrip
Working area
Runtime test status
Estimated parameters are displayed here
Motor Parameter Estimator
MOTOR PARAMETER ESTIMATOR
Open
Save
Motor Hardware Configurator
FILE
CONFIGURATION
Board
TIBoard
Motor
TEKNIC
HARDWARE SELECTION
Rs Estimation
Lq Estimation
Ld Estimation
Ke, J, and B Estimation
Estimate Parameters
TEST
Export
EXPORT
Algorithms
Rs Estimation
Ld Estimation
Lq Estimation
Ke, J, and B Estimation
Back EMF estimation time (s): 5
Back EMF estimation maximum speed limit (RPM): 3200
Speed lower limit for inertia estimation (RPM): 800
Iq reference current (pu): 0.3
Controller gains: Autocalculated
Proportional gain:
Integral gain:
Resistance (ohm): 0.2
D-axis inductance (H): 0.002
Restore the default values
Reset
System Messages
29-Apr-2024 16:23:33: Board TIBoard is selected. Select motor to enable …
29-Apr-2024 16:23:33: Motor TEKNIC is selected. Select algorithms to star…
Motor State
0 RPM
Fault Status
Communication
Results
Test Status: Not started
Status
ParameterName
Result
Phase resistance, Rs (ohm)
D-axis inductance, Ld (H)
Q-axis inductance, Lq (H)
Back-EMF constant, Ke (Vpk_LL/krpm)
Motor inertia, J (Kgm^2)
Friction constant, B (Nm/rad/s)
Last completed on: NA

Outcomes

Pain and requirements solved

App design workshop
Toolstrip app

- Various **workflows** of the PMSM estimation process are **scattered**, which makes it difficult for a new user to navigate the workflow
- No **control** over **individual test runs**
- In case of PMSM **test failure** user has to run all tests again

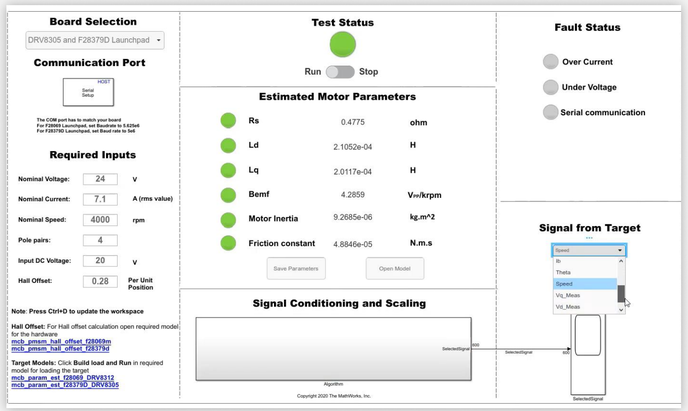


Outcomes

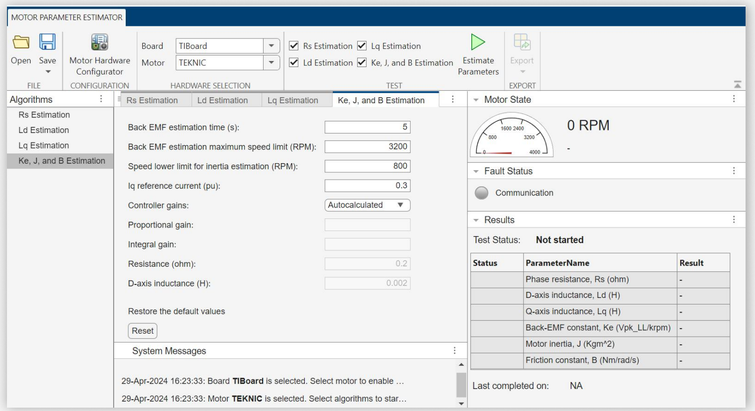
Pain and requirements solved

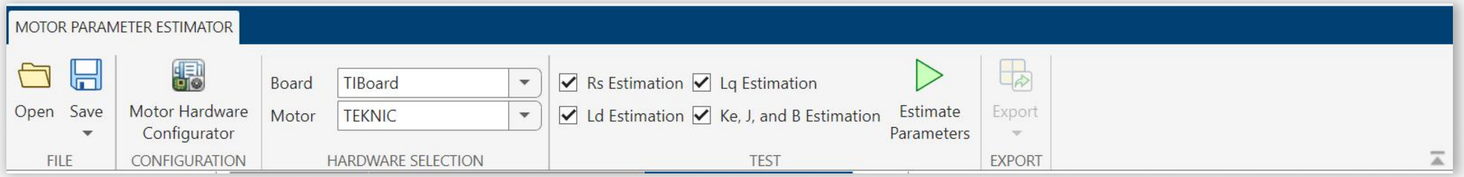
- The user does not have access to source code, so it is **impossible** for the user to **understand** the estimation process
- '**Session management**' with options to save progress and export results
- '**Real-time feedback**', including motor running status and completion warnings

Before



After





Prepare **Build** **Business use**

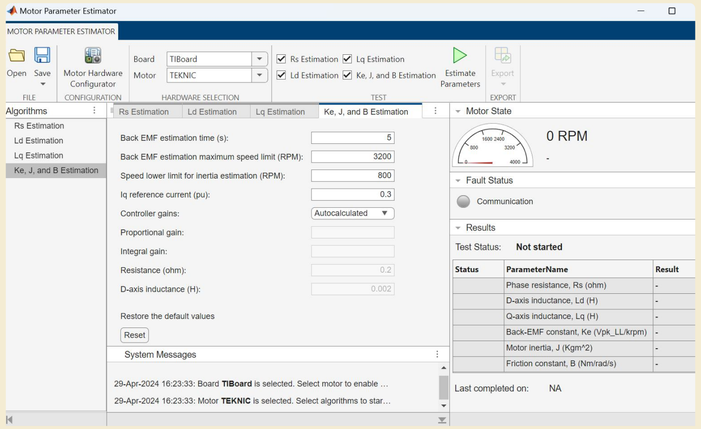


Impact

- Motor Control Block-set vertical is the fastest growing vertical of 2024 with a staggering 80+ percent growth.
- ~500% revenue growth over past three years
- ~30% growth in MCB toolbox enterprise users

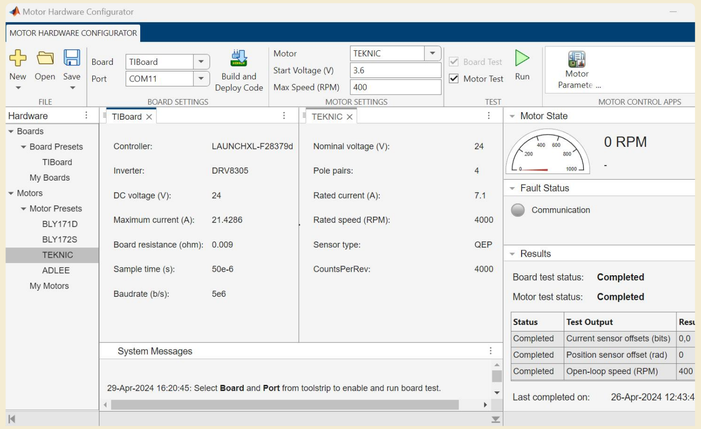
Future Scope

Integrated environment for all motor control workflows, along with AI integration



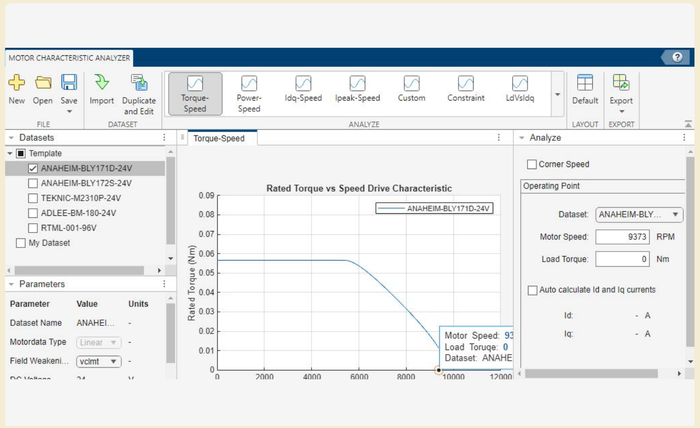
Motor Parameter Estimator

A motor control application to estimate asynchronous motor parameters.



Motor Hardware Configurator

Configuring hardware required for motor control in parameter estimation workflows



Motor Characteristic Analyzer

A quick and robust non-programmatic solution to plot motor control characteristics